COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:		**SECTION:**	
DATE PERFORMED:		**DATE SUBMITTED:**	
NAME:		**INSTRUCTOR:**	

ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

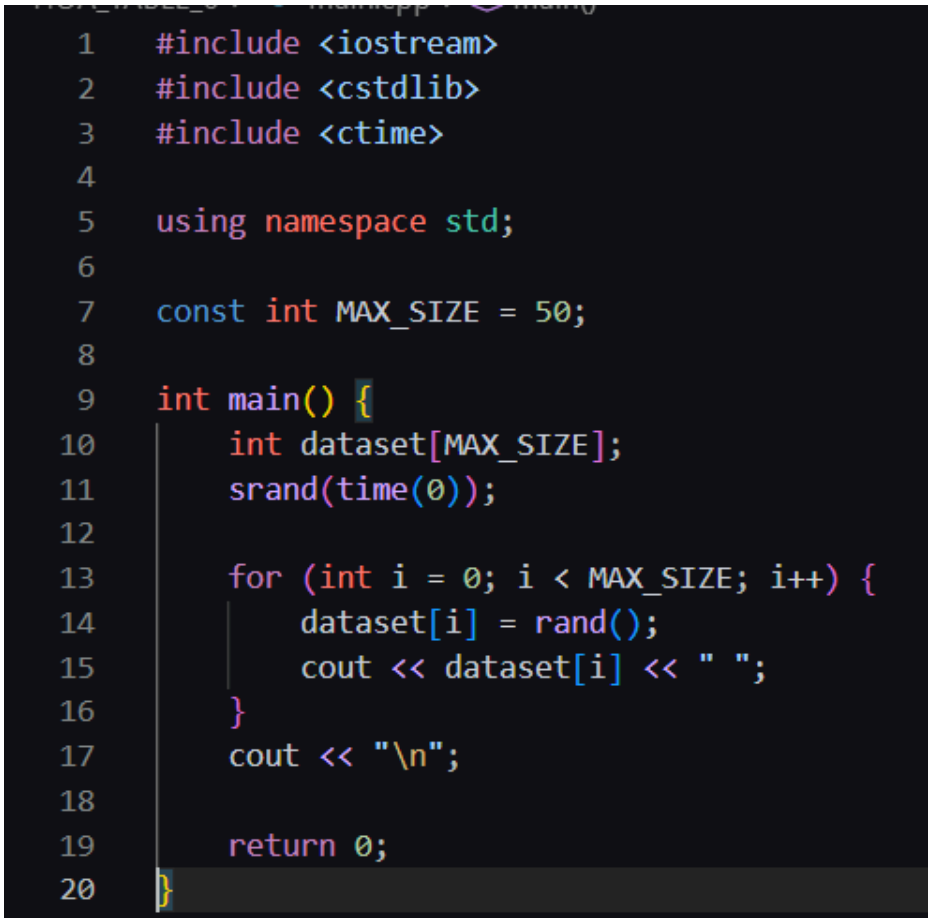
What procedures did you perform?

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

TABLE 6

Table 6-1. Data Generated and Observations.

SCREENSHOT	

```cpp
#include <iostream>
#include <cstdlib>
#include <ctime>

using namespace std;

const int MAX_SIZE = 50;

int main() {
    int dataset[MAX_SIZE];
    srand(time(0));

    for (int i = 0; i < MAX_SIZE; i++) {
        dataset[i] = rand();
        cout << dataset[i] << " ";
    }
    cout << "\n";

    return 0;
}
```

OBSERVATION	The `rand()` function fills the array with unsorted random numbers in contiguous memory to test search algorithms

Table 6-2a. Linear Search for Arrays

CODE	

```cpp
#include <iostream>
#include <cstdlib>
#include <ctime>
#include "searchingAlgo.h"

using namespace std;

const int MAX_SIZE = 50;

int main() {
    int dataset[MAX_SIZE];
    srand(time(0));

    for (int i = 0; i < MAX_SIZE; i++) {
        dataset[i] = rand();
    }

    int searchNum;
    cout << "Indicate the number to search: ";
    cin >> searchNum;

    linearSearchArr(MAX_SIZE, searchNum, dataset);

    return 0;
}
```

OUTPUT

```
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\CPE21S4 - SILVOZA>  ${env:DEBUGINFOD_URLS}=''; & 'c
ptools-1.34.0-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-rvsztd
' '--stderr=Microsoft-MIEngine-Error-2zmsmn2f.gks' '--pid=Microsoft-MIEngine-Pid-brly1ibr.xym' '--dbgExe=
14.0.0-r3\mingw64\bin\gdb.exe' '--interpreter=mi'
Indicate the number to search: 2the search is unsuccessful
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\CPE21S4 - SILVOZA> (Set-ExecutionPolicy -Scope Proc
th Silvoza\OneDrive\Pictures\Documents\CPE21S4 - SILVOZA\.venv\Scripts\Activate.ps1")
(CPE21S4 - SILVOZA) PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\CPE21S4 - SILVOZA>
(CPE21S4 - SILVOZA) PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\CPE21S4 - SILVOZA>
```

OBSERVATION	Linear search checks array elements one by one from start to end until the target number is found or missed.

Table 6-2b. Linear Search for Linked List

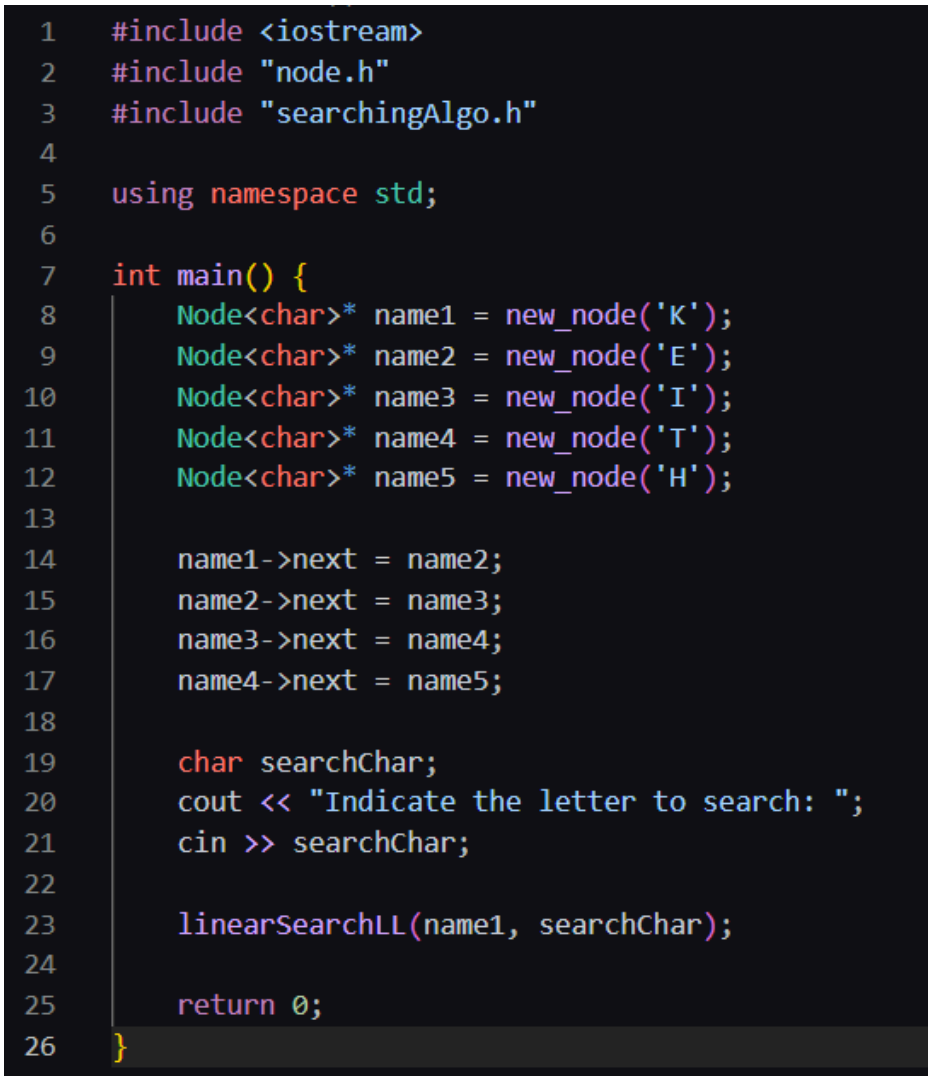
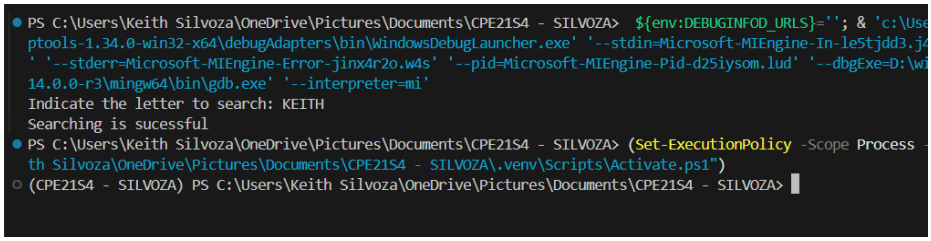
CODE	
OUTPUT	
OBSERVATION	The search traverses the linked list sequentially by following node pointers from head to tail to locate the character.

Table 6-3a. Binary Search for Arrays

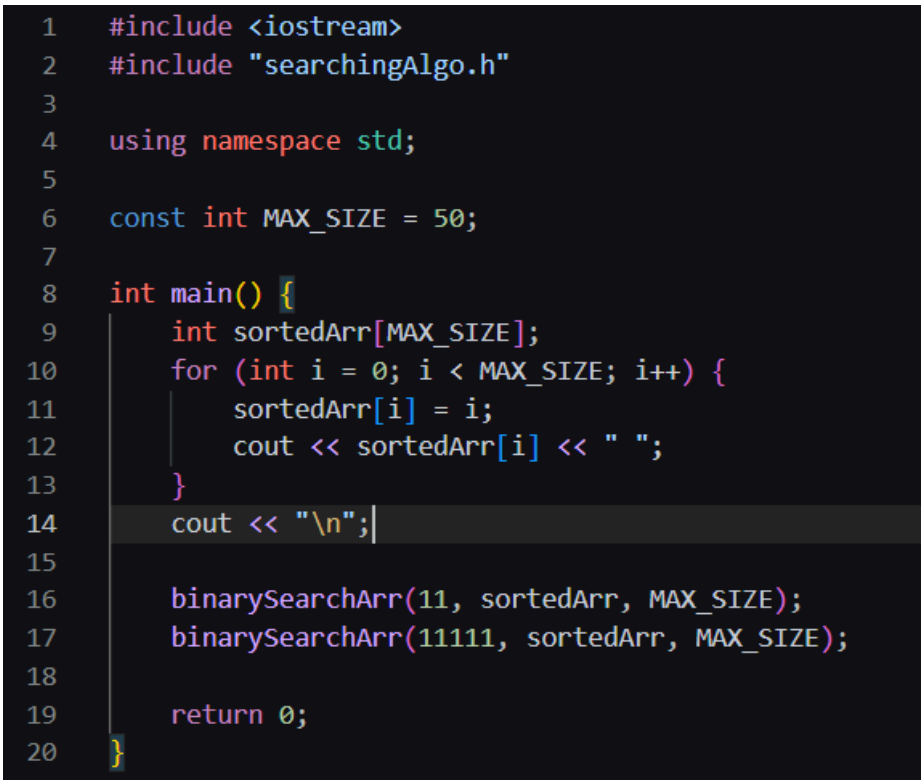
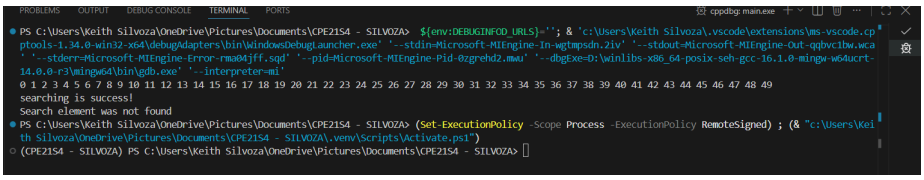
CODE	
OUTPUT	
OBSERVATION	Binary search splits the sorted array in half repeatedly, making searches much faster than linear scans.

Table 6-3b. Binary Search for Linked List

CODE

```cpp
#include <iostream>
#include "node.h"
#include "searchingAlgo.h"

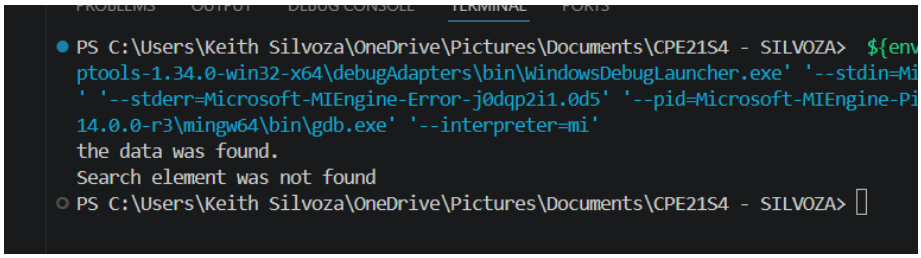
using namespace std;

int main() {
    Node<char>* s1 = new_node('A');
    Node<char>* s2 = new_node('B');
    Node<char>* s3 = new_node('C');
    Node<char>* s4 = new_node('D');
    Node<char>* s5 = new_node('E');
    Node<char>* s6 = new_node('F');
    Node<char>* s7 = new_node('G');

    s1->next = s2;
    s2->next = s3;
    s3->next = s4;
    s4->next = s5;
    s5->next = s6;
    s6->next = s7;

    binarySearchLL(s1, 'D');
    binarySearchLL(s1, 'Z');

    return 0;
}
```

OUTPUT	
OBSERVATION	Binary search on a linked list requires traversing pointers to find the middle node, adding pointer overhead compared to array indexing.

SUPPLEMENTARY ACTIVITY

```cpp
#include <iostream>
#include <string>
#include <limits>
#include "node.h"
#include "searchingAlgo.h"

using namespace std;

struct Student {
    int id;
    string name;

    bool operator==(const Student& other) const {
        return id == other.id;
    }
};

int main() {
    int searchID;
    int findVal;

    cout << "Enter Student ID to search: ";
    cin >> searchID;

    cin.ignore(numeric_limits<streamsize>::max(), '\n');

    cout << "Enter integer to binary search: ";
    cin >> findVal;

    cout << "\n--- RESULTS ---\n";

    const int STUDENT_COUNT = 5;
    Student students[STUDENT_COUNT] = {
        {101, "Alice"},
        {102, "Bob"},
        {103, "Charlie"},
        {104, "David"},
        {105, "Eve"}
    };

    Student targetStudent = {searchID, ""};
    cout << "Array Search: ";
    linearSearchArr(STUDENT_COUNT, targetStudent, students);

    Node<int>* head = new_node(10);
    Node<int>* n2 = new_node(20);
    Node<int>* n3 = new_node(30);
    Node<int>* n4 = new_node(40);
    Node<int>* n5 = new_node(50);

    head->next = n2;
    n2->next = n3;
    n3->next = n4;
    n4->next = n5;

    cout << "Linked List Binary Search: ";
    binarySearchLL(head, findVal);

    return 0;
}
```

RESULTS

```
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\CPE21S4 - SILVOZA>  ${env:DEBUGINFOD_URLS}=''; & 'c:\Users\Keith Silvoza\.vscode\extensions\ms-vscode.cpptools-1.34.0-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-dftkxv5x.vjw' '--stdout=Microsoft-MIEngine-Out-q1tqwjam.wz1' '--stderr=Microsoft-MIEngine-Error-ukpmmxsd.pe2' '--pid=Microsoft-MIEngine-Pid-2cln4cr1.zfe' '--dbgExe=D:\winlibs-x86_64-posix-seh-gcc-16.1.0-mingw-w64ucrt-14.0.0-r3\mingw64\bin\gdb.exe' '--interpreter=mi'
Enter Student ID to search: 103
Enter integer to binary search: 30

--- RESULTS ---
Array Search: Searching is sucessful
Linked List Binary Search: the data was found.
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\CPE21S4 - SILVOZA> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& "c:\Users\Keith Silvoza\OneDrive\Pictures\Documents\CPE21S4 - SILVOZA\.venv\Scripts\Activate.ps1")
(CPE21S4 - SILVOZA) PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\CPE21S4 - SILVOZA>
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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